# Lab 1: Make context diagram (level 0), level 1, level 2 data flow diagram and ER diagram for Bank Management System.

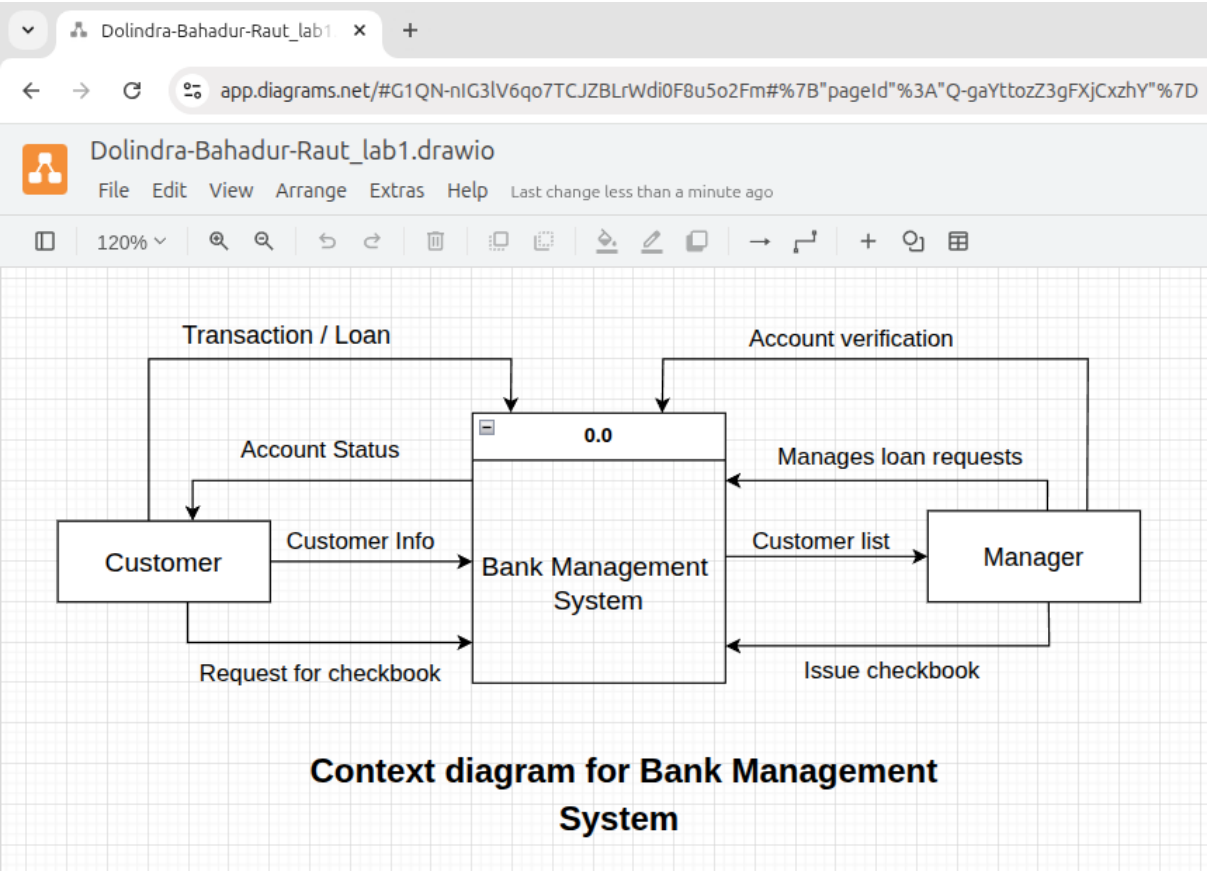
**Solution:**

**Level 0 Data flow diagram for Bank Management System:**

Level 0 DFD is also known as a context diagram. The context or level 0 DFD of Bank Management System represents the entire system as a single process and its relationship with external entities.

There are two external entities in this level 0 DFD for Bank Management System:

- Customer
- Manager



**Level 1 Data Flow Diagram for Bank Management System:**

Level 1 DFD for Bank Management System breaks the single process into the various sub process of the system. Here, the above single process has divided into 4 sub process in level 1 DFD. The sub processes are:

- **Account opening:**

  It maintains the record of all customer accounts and their personal details. It receives the details of customers in input and stores them in the database. Bank manager or staff verifies this application and after approval a new account is opened in the bank.

- **Manage Transaction:**

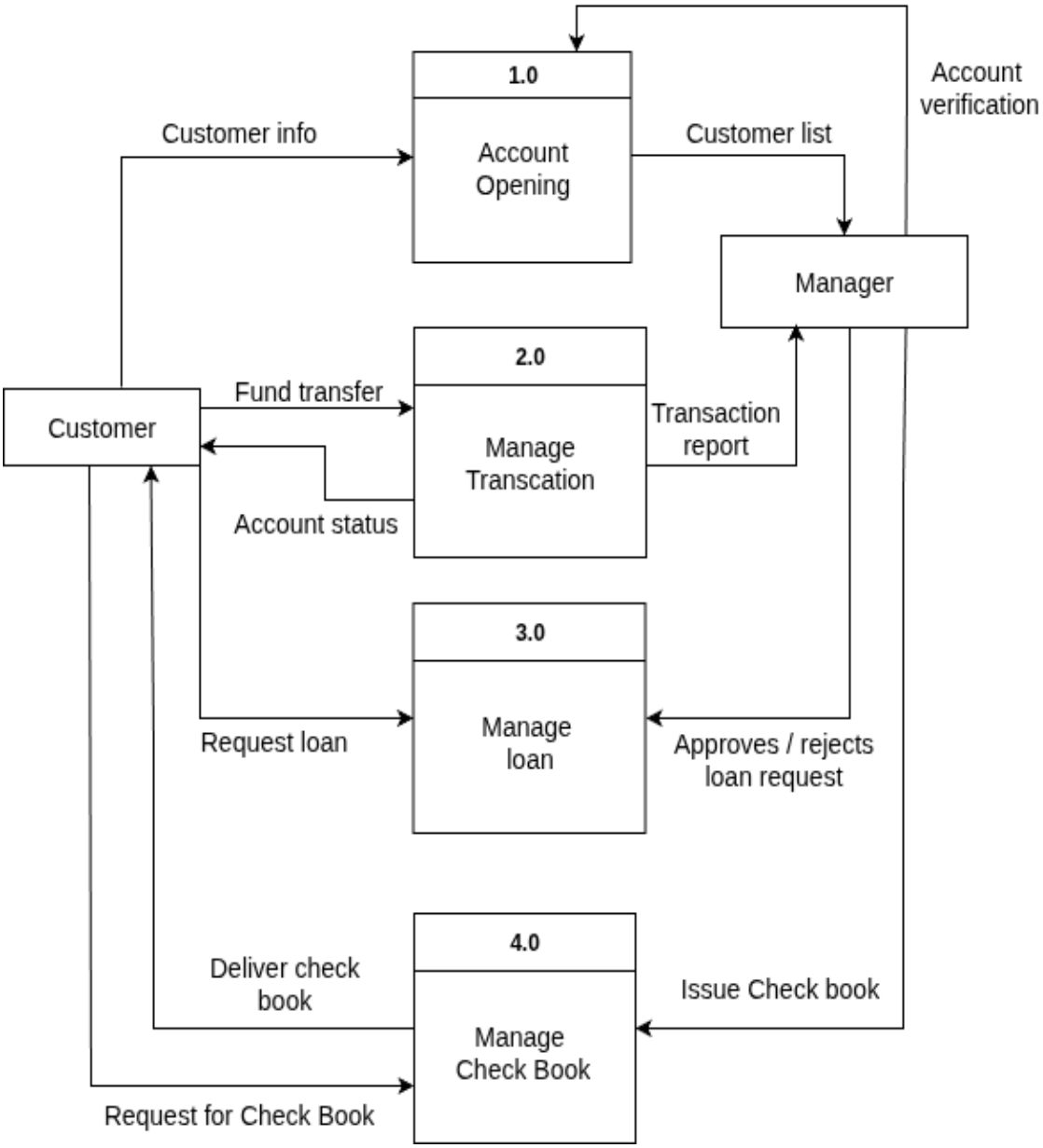
  This process keeps records of all the online transactions done by the account holders. For example: fund transfers, amount received. The transaction report data flows towards the bank manager.

- **Manage loan:**

  This process manages the loan request and confirmation. The customer can make request for loan. This request is handled by manage loan process. Bank manager can approve or reject this loan request.

- **Manage Checkbook:**

  Account holder can ask for new check book online. They can access bank website, access their account area and request for issuing new check book. This process receives the check book request and delivers check book at the customer's bank account. Bank manager issues the check book as per customer's requests.




Level 1 DFD for Bank Management System

**Level 2 Data Flow Diagram for Bank Management System:**

Level 2 DFD for online bank management system further breaks the level 1 process into more detailed sub-process and also gives information about data stores that maintain the record of the entire Bank Management System.

In level 2 DFD, the “Manage Transaction” process is further divided into 3 sub processes.

- **Add Beneficiary:** This sub process keeps record of beneficiary for future transaction.

- **Daily Transaction:**

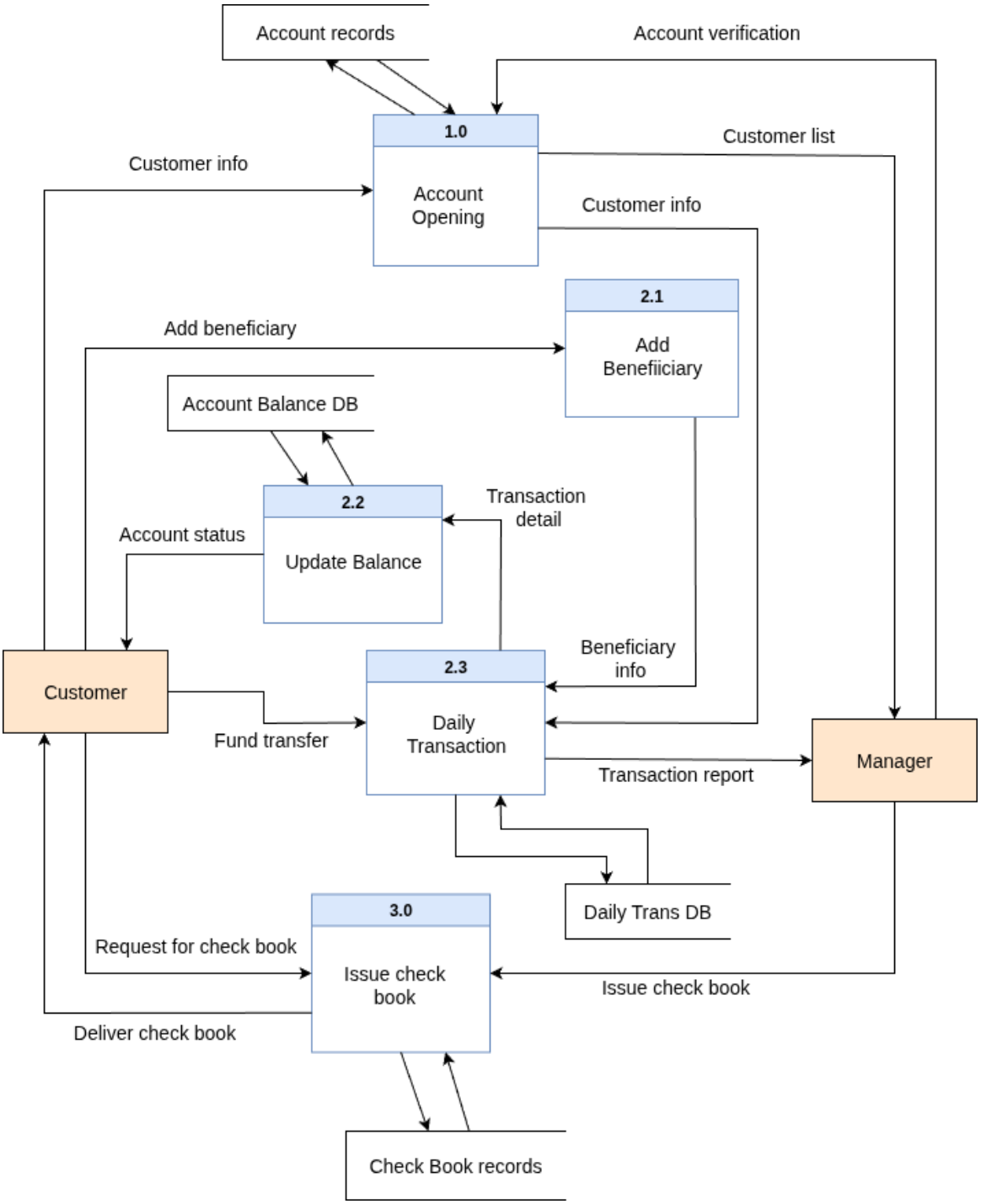
  It receives the customer’s details form the” Account opening” process which has the records of customer’s database. all the fund transfer requests done by the account holders.

  This process also takes beneficiary information from the “Add Beneficiary” process for processing fund transfer requests. Also, from this process, the transaction details go to the “Update Balance” process.

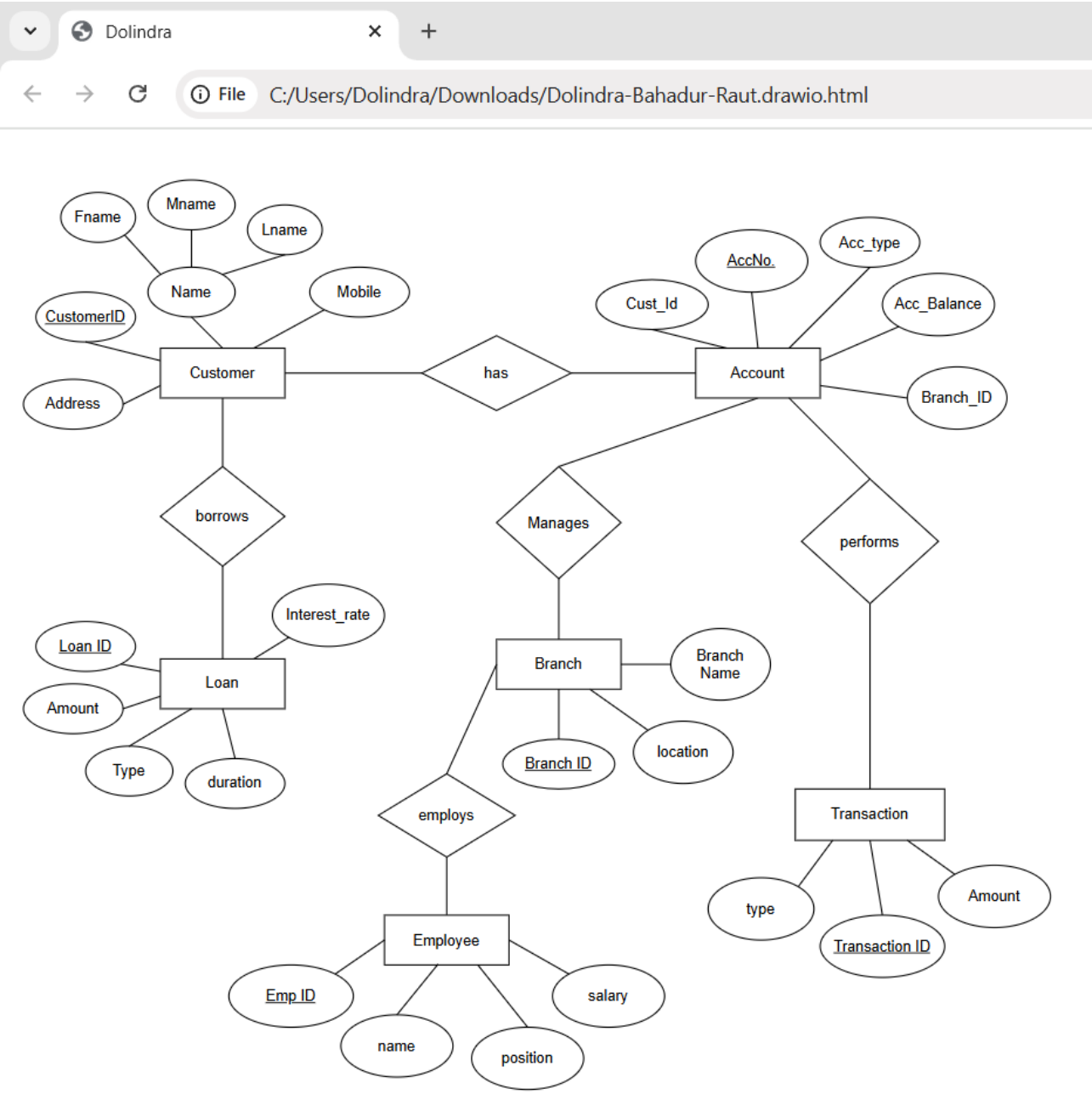
- **Update Balance:** After each transaction, this process updates the account balance. It keeps up to date information about customer’s account. Customer can view the account status. It receives the transaction details from the “Daily Transaction” process for updating the account balance.

Here, the following data stores have been used for DFD of Bank Management System:

- Account records
- Account Balance Database
- Daily transaction Database
- Check book records


Account records
Account verification
1.0
Account Opening
Customer list
Customer info
Customer info
2.1
Add Benefiiciary
Add beneficiary
Account Balance DB
2.2
Update Balance
Transaction detail
Account status
Beneficiary info
2.3
Daily Transaction
Customer
Fund transfer
Manager
Transaction report
Daily Trans DB
3.0
Issue check book
Request for check book
Issue check book
Deliver check book
Check Book records

## Entity-Relationship (ER) diagram for Bank Management System:



## Lab 2: Make level 0, level 1 and level 2 DFD (data flow diagram) and ER diagram for Library Management System.

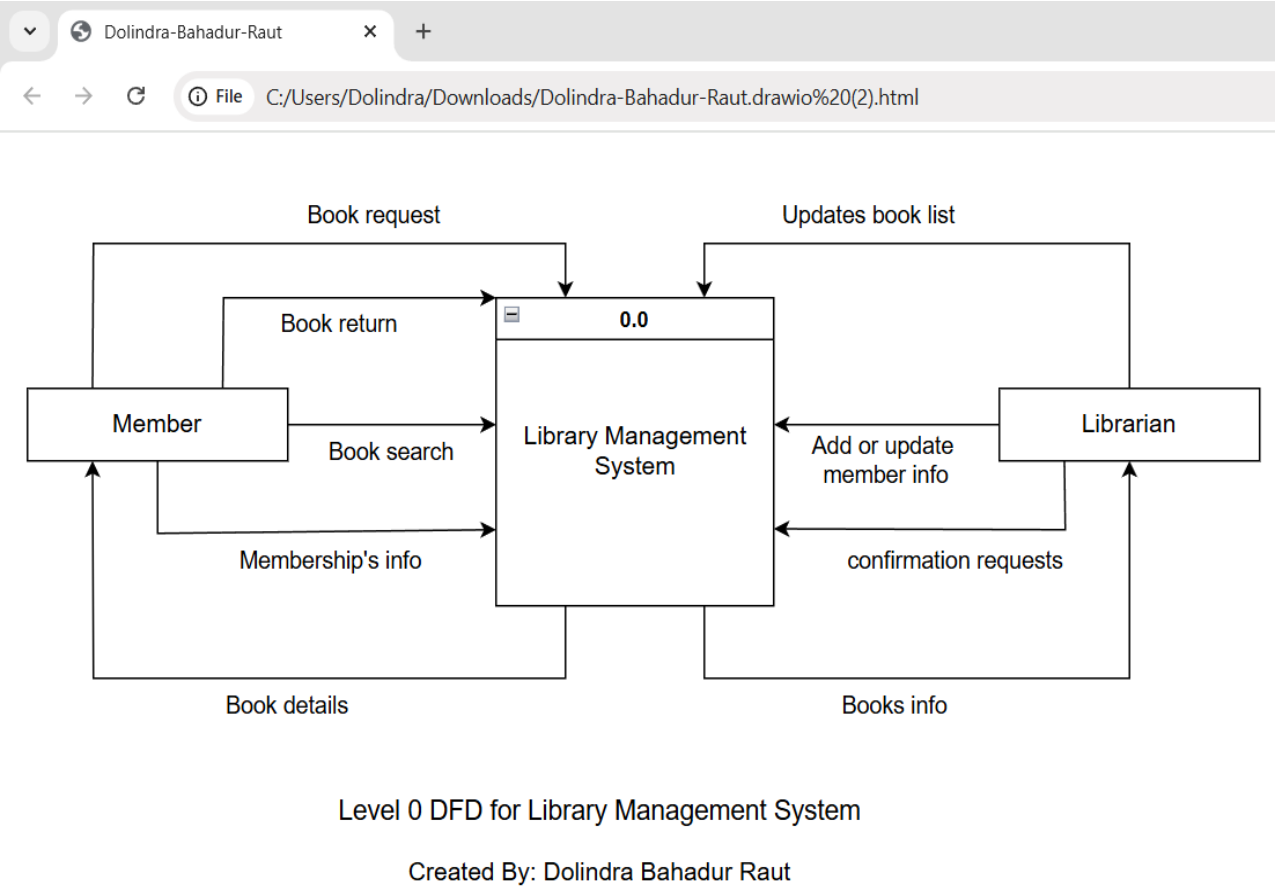
**Answer:**

**Level 0 Data flow diagram for Library Management System:**

The level 0 DFD of Library Management System represents the entire system as a single process and its relationship with external entities.

The below context diagram of Library Management System includes two external entities. They are:

1. **Member:** interacts with the system by making book requests, returning books, and searching for books.
2. **Librarian:** manages the library's operations and data.

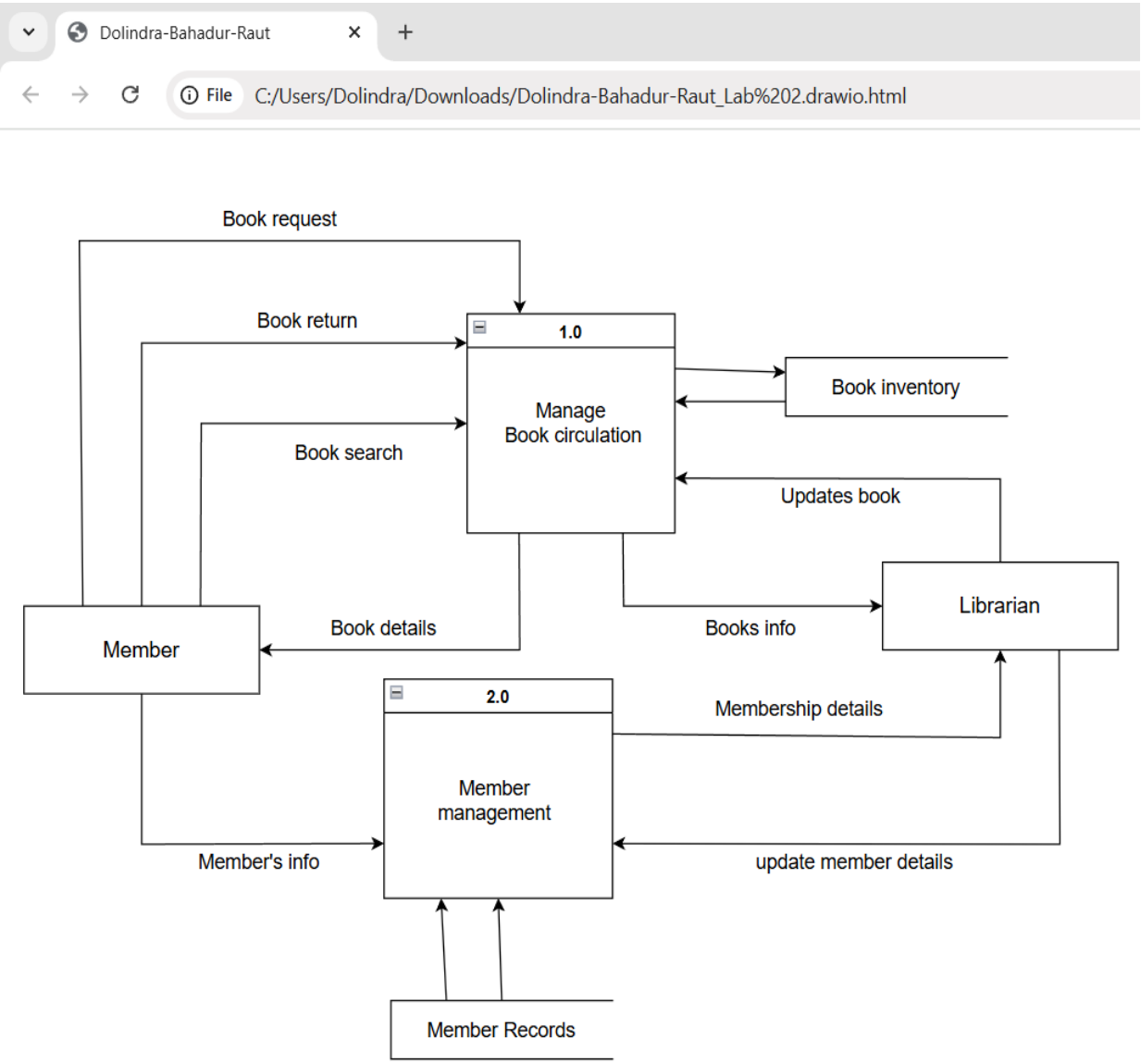


**Level 1 Data flow diagram for Library Management System:**

Here, in level 1 dfd, the single process has divided into two sub processes:

1.0. **Manage Book Circulation:** Handles book requests, returns, searches, and updates book inventory with the help of the librarian.

2.0. **Member management:** Maintains member records, processes membership details, and updates member information.



## Level 2 Data flow diagram for Library Management System:

In level 2 DFD, the “manage Book circulation” process is divided into 4 sub processes. They are:

### 1.1 Issue Book:

The “Issue Book” process handles the issuing of books to members. It receives a book issue request from the member, checks availability in the book inventory, and then issues the book. Sends confirmation to the member and updates the book inventory.

### 1.2 Search Book:

This process manages searching for books in the system. The member sends a search request, this process retrieves details from the book inventory, and sends the results back to the member.

### 1.3 Return Book:

It deals with returning books by members. It accepts returned books, updates the book inventory accordingly, and sends return confirmation to the member.

### 1.4 Update book info:

This process allows updating of book-related details (such as title, author, or availability). Updates are requested (often by the librarian), and the book inventory is modified. Sends update confirmation to the librarian.

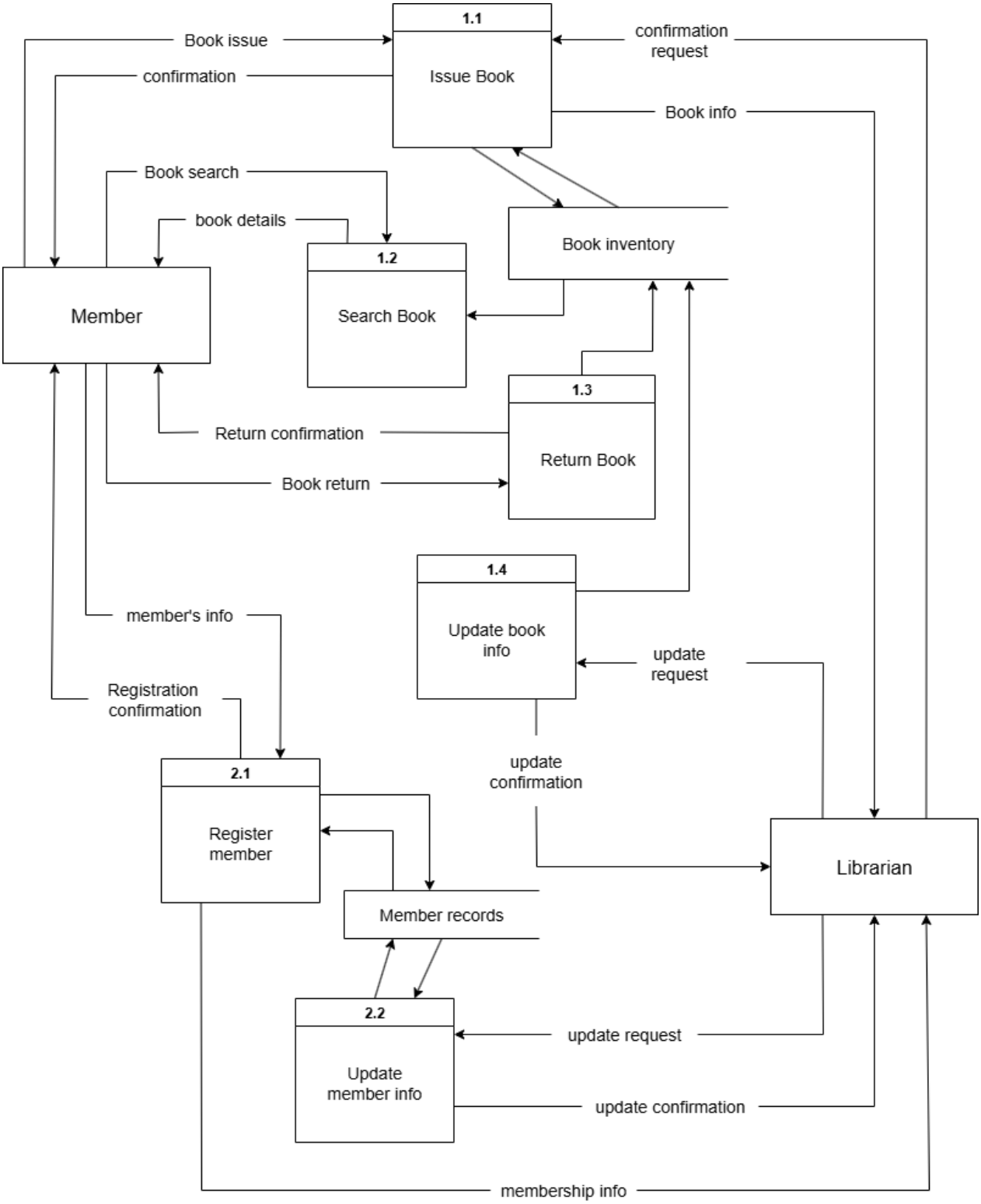
Also, the “member management” process is further divided into two sub processes:

**2.1 Register member:** records new member details into the system and confirms successful registration.

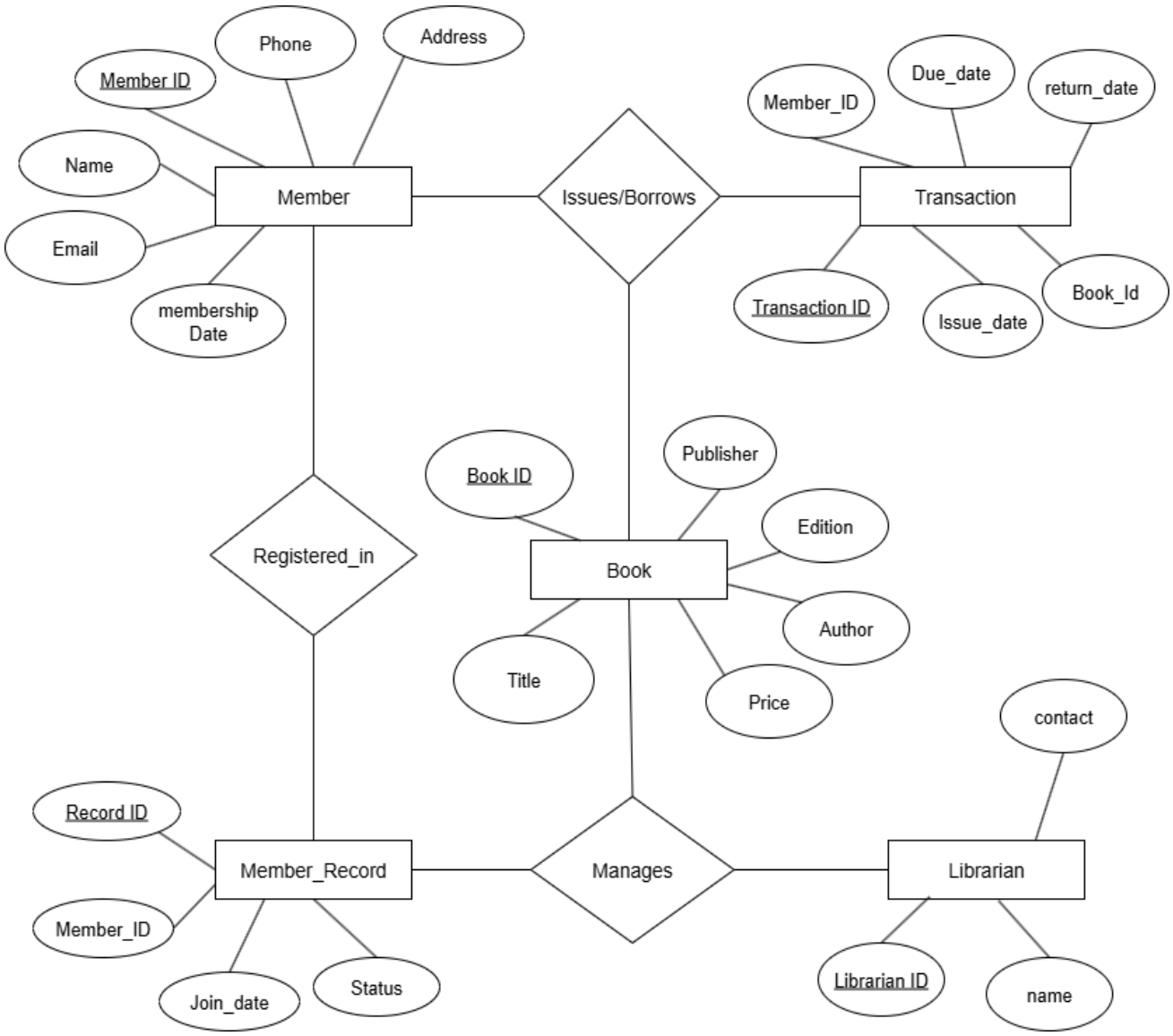
**2.2 Update member info:** modifies existing member records upon request and confirms the update to the librarian or member.

There are two data stores in this level 2 DFD of Library management system:

- Book inventory
- Member records


1.1
Issue Book
Book issue
confirmation
confirmation request
Book info
Book search
book details
Book inventory
1.2
Search Book
Member
1.3
Return Book
Return confirmation
Book return
1.4
Update book info
update request
member's info
Registration confirmation
update confirmation
2.1
Register member
Librarian
Member records
2.2
Update member info
update request
update confirmation
membership info

**ER diagram for Library Management System:**



## Lab 3: Make context diagram (level 0), level 1, level 2 data flow diagram and ER diagram for Online Employee Management System.

**Answer:**

An Employee Management System (EMS) is an information system designed to manage and automate the organizational processes related to employee data, attendance, leave, and salary. It helps organizations maintain accurate employee records, streamline HR operations, and improve efficiency by reducing manual paperwork. EMS acts as a centralized platform where all employee-related activities are stored, processed, and retrieved whenever needed.

**Level 0 Data flow diagram for Employee Management System:**

The Level 0 DFD, also known as the Context Diagram, represents the entire Employee Management System as a single process. It shows how the system interacts with the external entities and what data flows between them.

In this Level 0 DFD for the **Online Employee Management System,** there are two external entities:

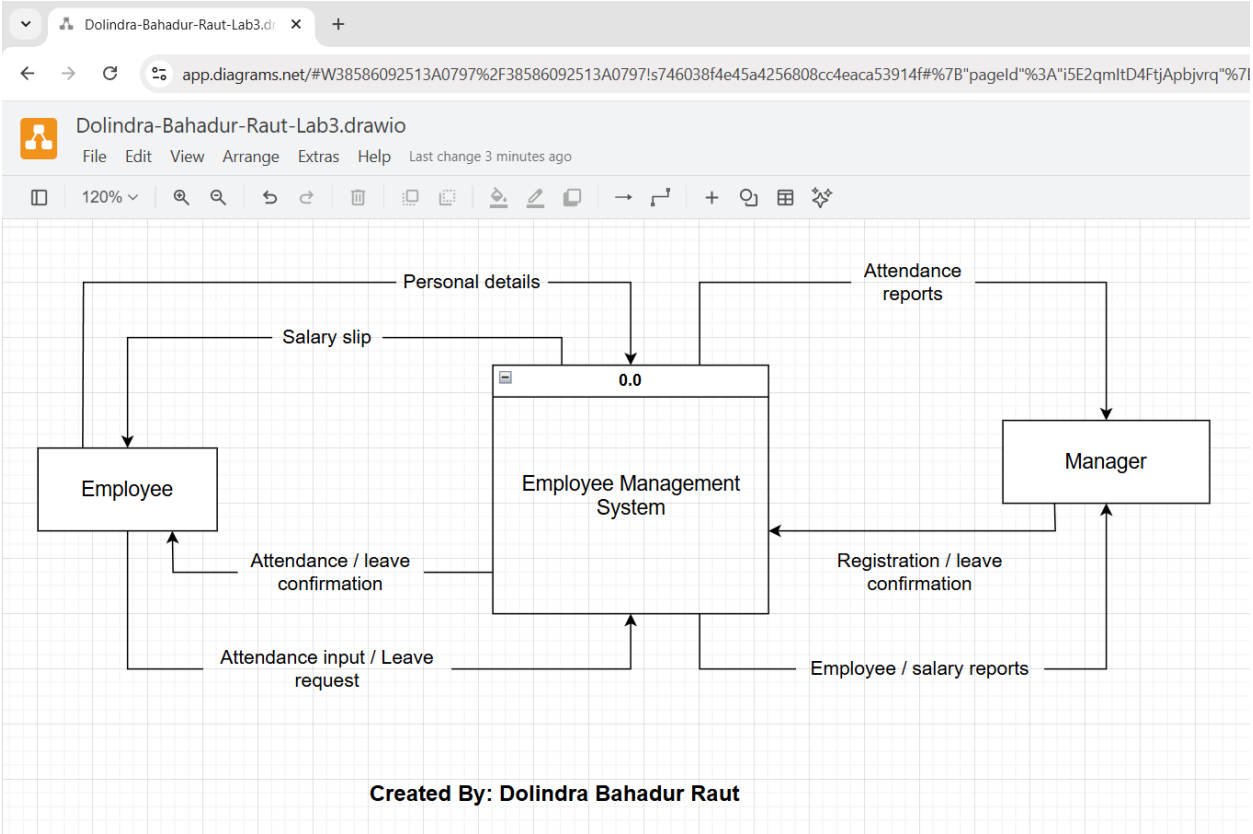
1. **Employee:**

   The employee interacts with the system by providing personal information, marking attendance, applying for leave, and viewing salary details.

2. **HR / Manager:**

   HR or Manager manages employee-related activities such as registration, updating employee information, verifying attendance, approving leave, and generating salary.

   These external entities exchange data with the system through well-defined data flows.



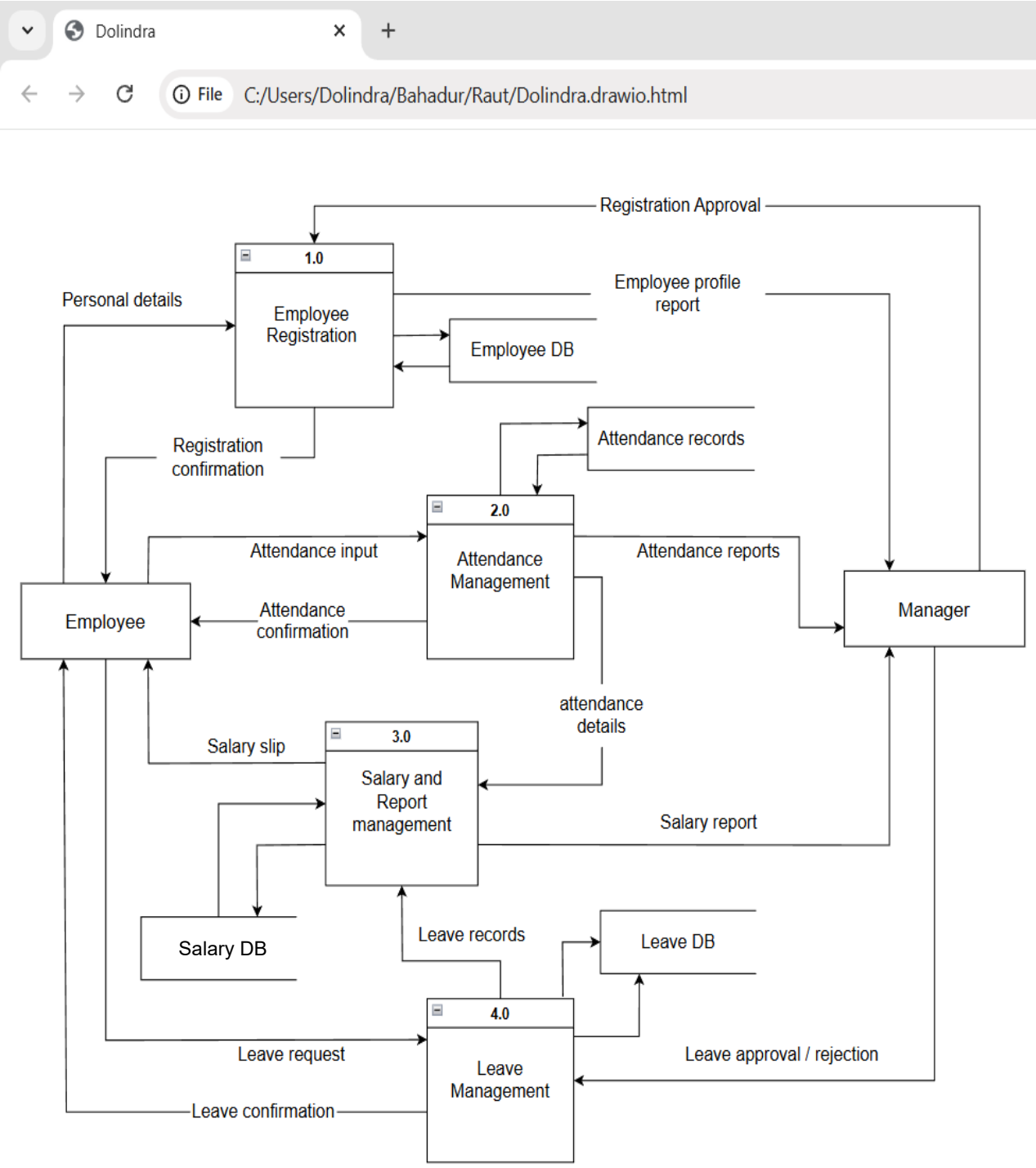

## Level 1 Data flow diagram for Employee Management System:

A Level-1 Data Flow Diagram (DFD) is the expanded version of a Level-0 DFD, where the main system is broken down into sub-processes for better understanding. It shows detailed data movement between processes, data stores, and external entities.

In this Employee Management System, Level-1 DFD decomposes the system into 4 major processes:

1. Employee Registration
2. Attendance Management
3. Salary and Report Management
4. Leave Management

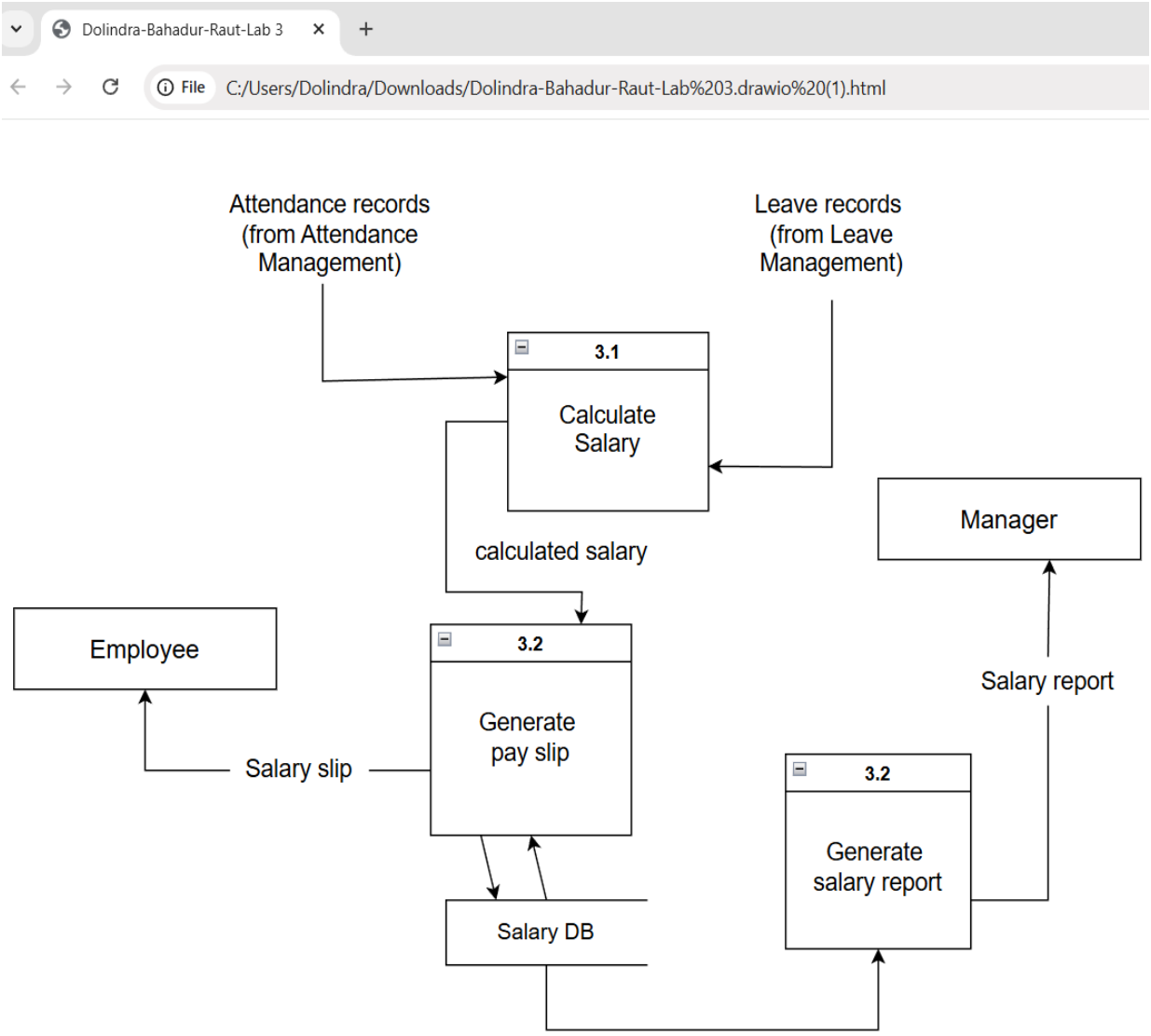
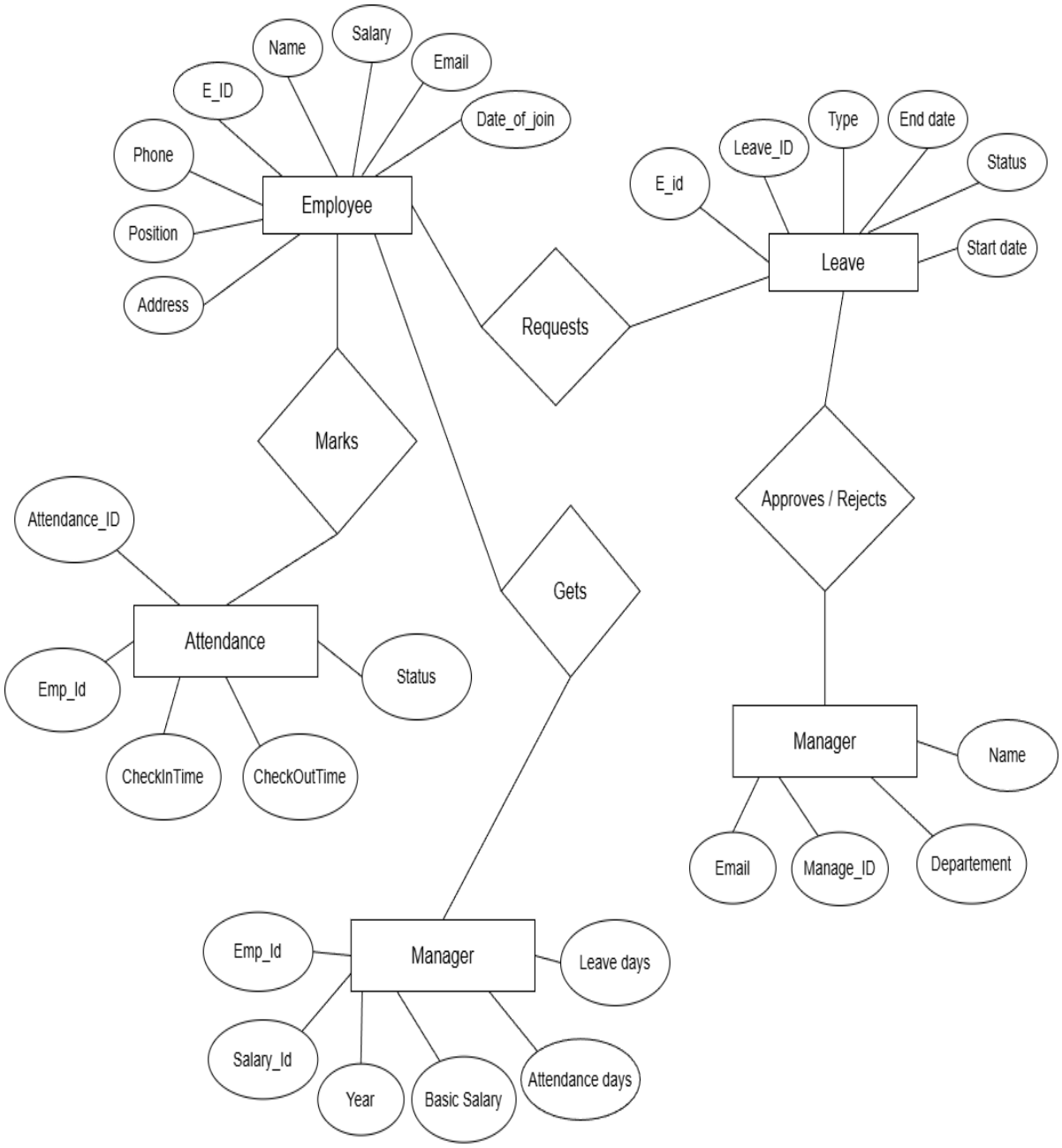
These processes interact with the Employee and Manager to handle data related to attendance, leave approval, payroll, and reports. It provides a clearer picture of how the system operates internally.


Dolindra
File
C:/Users/Dolindra/Bahadur/Raut/Dolindra.drawio.html
Registration Approval
1.0
Employee Registration
Personal details
Employee profile report
Employee DB
Registration confirmation
Attendance records
2.0
Attendance Management
Attendance input
Attendance reports
Employee
Attendance confirmation
Manager
attendance details
3.0
Salary and Report management
Salary slip
Salary report
Salary DB
Leave records
Leave DB
4.0
Leave Management
Leave request
Leave approval / rejection
Leave confirmation

**Level 2 Data flow diagram for Employee Management System:**

In level 2 DFD, the “Salary and report management” process is divided into 3 sub processes. They are:

- **Calculate Salary:** Computes each employee’s salary using attendance, leave, and pay details.
- **Generate Pay slip:** Creates individual pay slips from calculated salary.
- **Generate Salary Report:** Produces summary reports for management based on salary records and date range.



**ER diagram for Employee Management System:**